AI Explorer

Understand AI from the Ground Up

A comprehensive guide to AI — from foundational concepts and LLM comparisons to real-world products, code editors, community resources, and hands-on building guides.

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1. AI Concepts

25 foundational AI topics — from the broadest concepts to specialized techniques.

Artificial Intelligence (AI)

The broad field of building systems that can perform tasks requiring human-like intelligence — reasoning, learning, perception, and decision-making.

> **Key Insight:** AI is the umbrella — everything else on this page is a subset of AI.

Examples: Virtual assistants, Self-driving cars, Spam filters, Recommendation engines

Types: Narrow AI (current — task-specific), General AI (theoretical — human-level), Superintelligent AI (hypothetical).

Machine Learning (ML)

A subset of AI where algorithms learn patterns from data to make predictions — without being explicitly programmed for each scenario.

> **Key Insight:** Instead of writing rules, you give ML data and it finds the rules itself.

Three Types: Supervised Learning (labeled data), Unsupervised Learning (find hidden patterns), Semi/Self-supervised Learning (combination).

Deep Learning (DL)

A specialized subset of ML using multi-layered neural networks (often with billions of parameters) to learn complex patterns from massive datasets.

> **Key Insight:** Deep learning powers most modern AI breakthroughs — it's the engine behind the AI revolution.

Key Terms: Neural Network, Parameters, Backpropagation, GPU, Transformer architecture.

Generative AI (GenAI)

AI systems that create new content — text, images, video, audio, and code — based on patterns learned from training data.

> **Key Insight:** Traditional AI analyzes data. Generative AI creates new data. That's the paradigm shift.

Examples: ChatGPT writing essays, DALL-E creating images, Sora generating video, GitHub Copilot writing code.

Large Language Models (LLMs)

Deep learning models trained on massive text corpora that can understand and generate human language — the technology behind chatbots like ChatGPT and Claude.

> **Key Insight:** LLMs don't just retrieve information — they reason, summarize, translate, and create.

Key Terms: Token, Context Window, Transformer, RLHF, Inference, Open vs Closed models.

Computer Vision (CV)

AI that enables machines to interpret and extract meaningful information from images and video — from medical scans to autonomous driving.

Key architectures: CNNs, YOLO (object detection), Vision Transformers (ViT).

Natural Language Processing (NLP)

The branch of AI focused on enabling machines to understand, interpret, and generate human language — powering chatbots, translation, and search.

Core Tasks: Sentiment analysis, Named Entity Recognition, Translation, Summarization, Question Answering.

Reinforcement Learning (RL)

AI that learns through trial and error — an agent interacts with an environment, receiving rewards or penalties, to discover optimal strategies.

Key: RLHF (RL from Human Feedback) is the technique used to align ChatGPT and Claude with human preferences.

AI Agents

Autonomous AI systems that can plan, reason, use tools, and take multi-step actions to accomplish goals. Follow an agentic loop: Observe → Think → Act → Evaluate → Repeat.

Examples: Claude Code, AutoGPT, Devin, CrewAI multi-agent systems.

Model Context Protocol (MCP)

An open standard created by Anthropic that defines how AI applications communicate with external tools and data sources. One universal protocol — any AI app that supports MCP can connect to any MCP server.

Architecture: MCP Host (AI app) → MCP Client → MCP Server (exposes tools). Primitives: Tools, Resources, Prompts.

Retrieval-Augmented Generation (RAG)

A technique that retrieves relevant documents from a knowledge base and feeds them to an LLM as context, enabling accurate answers grounded in specific data.

Pipeline: Split docs → Embed → Store in vector DB → Retrieve relevant chunks → Augment prompt → Generate answer.

Fine-Tuning

Adapting a pre-trained model to a specific task or domain by continuing training on a smaller, specialized dataset.

Methods: Full fine-tuning (expensive), LoRA (efficient, ~0.5% params), QLoRA (4-bit quantization + LoRA), API-based (no GPU needed).

Embeddings

Converting text, images, or other data into dense numerical vectors that capture semantic meaning. Similar items have similar embeddings.

Models: OpenAI text-embedding-3-small, Google text-embedding-004, nomic-embed-text (local).

Vector Databases

Specialized databases optimized for storing, indexing, and searching embedding vectors using similarity metrics like cosine similarity.

Examples: Pinecone, Weaviate, Chroma, Milvus, pgvector.

Prompt Engineering

The practice of designing effective inputs (prompts) to get optimal outputs from AI models. Better prompts produce dramatically better results.

Techniques: Few-shot, Chain-of-thought, System prompts, Role-playing, Structured output.

Transformers

The neural network architecture behind virtually all modern LLMs. Uses self-attention to understand relationships between all parts of the input simultaneously.

Key Paper: "Attention Is All You Need" (2017) — replaced recurrence with self-attention.

Transfer Learning

Using knowledge learned from one task to improve performance on a different, related task. Pre-training on general data, then specializing.

Key Insight: This is why you can fine-tune a model trained on the internet for your specific domain.

Diffusion Models

Generative models that learn to create images by gradually removing noise. The foundation behind DALL-E, Midjourney, and Stable Diffusion.

Process: Add noise to images → Train model to reverse the noise → Generate new images from pure noise.

MLOps

The practice of deploying, monitoring, and maintaining ML models in production. The "DevOps" of machine learning.

Key concerns: Model versioning, data drift detection, A/B testing, monitoring, retraining pipelines.

Multimodal AI

AI systems that can process and generate multiple types of content — text, images, audio, video — within a single model.

Examples: GPT-5.2 (text + vision + audio), Gemini 2.5 (text + image + video), Claude Opus 4.6 (text + vision).

Federated Learning

Training AI models across decentralized devices without sharing raw data. Each device trains locally and only shares model updates.

Use cases: Healthcare (HIPAA compliance), mobile keyboards (next-word prediction), financial fraud detection.

Responsible AI

The practice of developing and deploying AI systems that are fair, transparent, accountable, and safe.

Concerns: Bias, hallucination, privacy, explainability, environmental impact, job displacement.

Knowledge Graphs

Structured representations of real-world entities and their relationships, enabling AI to reason about connected information.

Examples: Google Knowledge Graph, Wikidata, enterprise knowledge management systems.

Synthetic Data

Artificially generated data that mimics real-world data, used when real data is scarce, expensive, or privacy-sensitive.

Uses: Training data augmentation, testing, privacy preservation, bias reduction.

Edge AI

Running AI models directly on devices (phones, cameras, IoT) rather than in the cloud, enabling faster responses and offline operation.

Examples: On-device speech recognition, real-time video analysis, autonomous vehicle processing.

2. LLM Model Rubric

27 models from 12+ providers — compared by capabilities, pricing, and use cases.

Model	Provider	Context	Pricing Tier	Category	Docs
GPT-5.2	OpenAI	200K	Premium	Frontier	Link
GPT-4.1	OpenAI	1M	Standard	Workhorse	Link
GPT-4.1 mini	OpenAI	1M	Budget	Lightweight	Link
o3	OpenAI	200K	Premium	Reasoning	Link
o4-mini	OpenAI	200K	Standard	Reasoning	Link
Claude Opus 4.6	Anthropic	200K	Premium	Frontier	Link
Claude Sonnet 4.6	Anthropic	200K	Standard	Workhorse	Link
Claude Haiku 4.5	Anthropic	200K	Budget	Lightweight	Link
Gemini 2.5 Pro	Google	1M	Premium	Frontier	Link
Gemini 2.5 Flash	Google	1M	Budget	Workhorse	Link
Gemini 2.5 Flash-Lite	Google	1M	Budget	Lightweight	Link
Llama 4 Maverick	Meta	1M	Free (open)	Frontier	Link
Llama 4 Scout	Meta	10M	Free (open)	Workhorse	Link
Llama 3.3 70B	Meta	128K	Free (open)	Community	Link
Mistral Large 2	Mistral	128K	Standard	Frontier	Link
Mistral Small 3.1	Mistral	128K	Budget	Lightweight	Link
Codestral	Mistral	256K	Standard	Code	Link
Command R+	Cohere	128K	Standard	RAG	Link
Grok-4	xAI	256K	Premium	Frontier	Link
Grok-3 mini	xAI	128K	Standard	Reasoning	Link
DeepSeek-V3	DeepSeek	128K	Budget	Open	Link
DeepSeek-R1	DeepSeek	128K	Budget	Reasoning	Link
Phi-4 Reasoning	Microsoft	32K	Free (open)	Small	Link
Qwen3 235B	Alibaba	128K	Budget	Open	Link
Qwen3 32B	Alibaba	128K	Free (open)	Community	Link
Amazon Nova Pro	Amazon	300K	Budget	Enterprise	Link

Jamba 1.5 Large	AI21 Labs	256K	Standard	Enterprise	Link

3. Model Training & AI History

AI Timeline (1950–2025)

Year	Event	Significance
1950	**The Turing Test**	Alan Turing publishes "Computing Machinery and Intelligence," proposing machines could think.
1956	**Dartmouth Conference**	John McCarthy coins "artificial intelligence" — the official birth of AI as a field.
1958	**The Perceptron**	Frank Rosenblatt builds the first artificial neural network capable of learning.
1974	**First AI Winter**	Funding collapses after Lighthill Report criticizes AI progress.
1980	**Expert Systems Rise**	Rule-based expert systems bring commercial success, then lead to second AI winter.
1997	**Deep Blue Beats Kasparov**	IBM's Deep Blue defeats world chess champion using brute-force search.
2012	**AlexNet & Deep Learning**	Deep CNN wins ImageNet by huge margin, igniting the deep learning revolution.
2017	**Attention Is All You Need**	Transformer architecture paper — foundation for GPT, BERT, and all modern LLMs.
2019	**GPT-2 & Scaling Begins**	OpenAI shows scaling up transformers produces surprisingly coherent text.
2022	**ChatGPT Changes Everything**	ChatGPT reaches 100M users in two months, making AI accessible to all.
2023	**The Frontier Race**	GPT-4, Claude 2, Gemini, Llama 2, Mistral launch in rapid succession.
2024	**Reasoning & Agents**	Chain-of-thought reasoning (o1, Claude 3.5), tool use, and agentic capabilities emerge.
2025	**The Agent Era**	Claude Opus 4, GPT-5, Gemini 2.5 push boundaries with deep reasoning and multi-step agents.

Training Technologies

RLHF (Reinforcement Learning from Human Feedback) OpenAI

Human raters rank model outputs; a reward model learns preferences, then the LLM is optimized via PPO to maximize that reward signal. This is what made ChatGPT feel conversational vs raw GPT-3.

Constitutional AI (CAI) Anthropic

The model critiques its own outputs against a set of principles ("constitution"), reducing reliance on human labelers. Uses RLAIF (RL from AI Feedback) for the reward model.

Mechanistic Interpretability Anthropic

Reverse-engineering how neural networks work by identifying interpretable features, circuits, and computational structures inside trained models.

Scaling Laws & Compute-Optimal Training OpenAI

Predictable relationships between model size, dataset size, compute budget, and performance. Chinchilla showed most models were under-trained on too little data.

Pathways & Multi-Task Training Google

One model handles thousands of tasks across modalities. Sparse activation means only a fraction of parameters are used per input. Foundation for Gemini.

Knowledge Distillation Google

A large "teacher" model's knowledge is compressed into a smaller, faster "student" model. Used for Gemini Flash and Nano variants.

Open-Weight Training & Community Meta

Meta releases model weights publicly. Quantization techniques (GGUF, GPTQ, AWQ) make large models run on consumer hardware.

FSDP (Fully Sharded Data Parallel) Meta

Shards model parameters, gradients, and optimizer states across GPUs. Scales to thousands of GPUs with near-linear efficiency. Now a core PyTorch feature.

Colossus Supercluster Training xAI

xAI built Colossus with 100,000+ H100 GPUs. Combined with real-time data from X for training data freshness.

Mixture of Experts (MoE) Mistral

A routing mechanism sends each token to only a subset of specialized "expert" sub-networks. Mixtral 8x7B has 47B total params but uses ~13B per forward pass.

Pros & Cons of LLMs

Pros

- **Unprecedented Productivity** — 10x output across knowledge work.

Cons

- **Hallucination** — Confidently generates incorrect information.

- **Democratized Knowledge** — Expert-level explanations for anyone.
- **Accelerated Development** — AI coding assistants ship code faster.
- **Breaking Language Barriers** — Real-time translation.
- **Scientific Discovery** — Literature review, hypothesis generation.
- **Accessibility** — Voice interfaces, content simplification.
- **Bias Amplification** — Inherits biases from training data.
- **Energy Consumption** — Massive compute requirements.
- **Concentration of Power** — Few companies train frontier models.
- **Copyright Concerns** — Training data legal uncertainty.
- **Job Displacement** — Automation of knowledge work.

Individual Fine-Tuning Guide

What is Fine-Tuning? Takes a pre-trained model and continues training on your specific dataset. Much cheaper than training from scratch — hours instead of months.

Method	VRAM	Params Trained	Speed	Cost
Full Fine-Tuning	160+ GB	100%	Slow	$$$
LoRA	24-48 GB	~0.5%	Fast	$
QLoRA	12-24 GB	~0.5%	Fast	$

Approach	When to Use	Examples Needed	Best For
Prompt Engineering	Quick results, base model almost works	<50	Quick iteration
RAG	Need specific docs, knowledge changes	N/A (docs)	Knowledge-heavy tasks
Fine-Tuning	Consistent style/format, quality ceiling hit	500+	Style & format

4. Prompt Engineering

18 techniques across 5 categories for getting better results from AI models.

Quick Tips

- **Be Specific, Not Vague** — Replace generic requests with precise instructions (format, length, tone, audience).
- **Give Context & Role** — Set the scene: "You are a senior Python developer reviewing code for a junior engineer."
- **Show, Don't Just Tell** — Include examples of what you want. One concrete example is worth a hundred words.
- **Break Down Complex Tasks** — Split big tasks into sequential steps.
- **Iterate and Refine** — Treat prompting as a conversation, not a one-shot.

Core Techniques

Zero-Shot Prompting

Ask the model to perform a task without any examples. Works for simple, well-defined tasks where the model has strong training data.

Few-Shot Prompting

Provide 2-5 examples of desired input-output pairs before your actual request. Dramatically improves consistency and formatting.

Chain-of-Thought (CoT)

Ask the model to "think step by step." Forces explicit reasoning that catches errors and produces better answers for math, logic, and analysis.

System Prompts

Set persistent instructions that define the AI's role, tone, constraints, and output format across the entire conversation.

Role-Based Prompting

Assign the model a specific persona: "You are a security auditor." Changes tone, vocabulary, and focus areas.

Structured Output

Request output in a specific format (JSON, Markdown, tables). Use schema definitions for reliable machine-readable responses.

Advanced Techniques

Tree of Thoughts

Explore multiple reasoning paths simultaneously and evaluate which approach leads to the best answer. For complex problems with multiple valid approaches.

Self-Consistency

Generate multiple responses to the same prompt and take the majority answer. Reduces variance and improves reliability on reasoning tasks.

ReAct (Reason + Act)

Interleave reasoning ("I need to find X") with actions ("search for X") and observations ("I found Y"). The foundation for AI agents.

Meta-Prompting

Ask the AI to generate or improve prompts. "What questions should I ask you to get the best result for X?"

Constrained Generation

Set explicit boundaries: word limits, vocabulary restrictions, format requirements. Prevents rambling and ensures output meets specifications.

Prompt Chaining

Break complex tasks into sequential prompts where each output feeds into the next. Research → Outline → Draft → Edit.

5. AI Products Explorer

50 companies building the AI ecosystem — from model providers to developer tools to creative platforms.

Company	Key Products	Category
OpenAI	ChatGPT, GPT-5.2, DALL-E, Sora	Model Provider
Anthropic	Claude, Claude Code, Agent SDK	Model Provider
Google	Gemini, Vertex AI, NotebookLM	Model Provider
Microsoft	Copilot, Azure AI, Phi models	Platform
Meta	Llama, PyTorch, AI Studio	Open Source
AWS	Bedrock, SageMaker, Nova	Cloud
Midjourney	Midjourney image generation	Image Gen
Stability AI	Stable Diffusion, SDXL	Image Gen
Hugging Face	Model Hub, Transformers, Spaces	Platform
Perplexity	AI-powered search engine	Search
Runway	Gen-3 video generation	Video Gen
ElevenLabs	Voice synthesis, dubbing	Audio
Databricks	Lakehouse AI, MLflow, DBRX	Data + ML
Cohere	Command R+, Embed, RAG	Enterprise AI
NVIDIA	GPUs, CUDA, NeMo, TensorRT	Hardware + SW
LangChain	LangChain, LangSmith, LangGraph	Framework
LlamaIndex	Data framework for LLM apps	Framework
Pinecone	Vector database (managed)	Vector DB
Weaviate	Open-source vector database	Vector DB
CrewAI	Multi-agent orchestration	Agents
AutoGen	Microsoft multi-agent framework	Agents
Scale AI	Data labeling, RLHF, evaluation	Data
Weights & Biases	Experiment tracking, MLOps	MLOps
Snowflake	Cortex AI, Arctic, data cloud	Data + AI
Vercel	AI SDK, v0 (UI gen)	Frontend
GitHub	Copilot, Copilot Workspace	Code

Cursor	AI-native code editor	Code
Jasper	AI marketing content	Marketing
Notion	Notion AI assistant	Productivity
Salesforce	Einstein AI, Agentforce	Enterprise CRM
IBM	watsonx, Granite models	Enterprise
Palantir	AIP (AI Platform)	Enterprise
Mistral AI	Mistral Large, Codestral, Le Chat	Model Provider
Groq	LPU inference (ultra-fast)	Inference
Replicate	Model hosting & API	Inference
Descript	AI video/audio editing	Media
Synthesia	AI video avatars	Video
Grammarly	AI writing assistant	Writing
Together AI	Open-source model hosting	Inference
Modal	Serverless GPU compute	Infrastructure
xAI	Grok-4, Colossus cluster	Model Provider
DeepSeek	DeepSeek-V3, R1 (open)	Model Provider
Canva	Magic Design, AI image gen	Design
Adobe	Firefly, Sensei AI	Creative
Figma	AI design tools	Design
Suno	AI music generation	Music
Replit	AI coding & deployment	Code
Zapier	AI automation workflows	Automation
Speechify	AI text-to-speech	Audio

6. AI Tools & Protocols

Understand the building blocks of modern AI systems — what agents are, how they connect to tools through MCP, what APIs do, and how it all fits together.

Quick Reference: How It All Fits Together

> **1.** You ask an AI Agent to do something
> **2.** The Agent reasons about what tools it needs (Function Calling)
> **3.** It discovers available tools via MCP (tools/list)
> **4.** It calls the right tool, which hits an API endpoint
> **5.** The API authenticates the request using an API Key
> **6.** The result flows back through the chain to the Agent
> **7.** The Agent uses the result to continue or complete your task

AI Agents

Tagline: AI that acts, not just responds

An AI agent is an autonomous system that can plan, reason, use tools, and take multi-step actions. Agents follow an "agentic loop": Observe → Think → Act → Evaluate → Repeat.

Examples: Claude Code (Anthropic's CLI agent), Devin, AutoGPT, CrewAI

Analogy: A chatbot is a reference librarian — ask a question, get an answer. An agent is an executive assistant — say "plan my trip to Tokyo" and they book flights, reserve hotels, and send the itinerary.

Model Context Protocol (MCP)

Tagline: USB-C for AI — one standard to connect everything

MCP is an open standard created by Anthropic that defines how AI apps communicate with external tools. Architecture: MCP Host → MCP Client → MCP Server. Three primitives: Tools (actions), Resources (data), Prompts (templates).

Examples: Claude Desktop, GitHub MCP Server, Database Servers, Slack/Calendar integrations

APIs

Tagline: The plumbing that connects all software

An API (Application Programming Interface) is how software talks to other software. Common styles: REST (most popular), GraphQL (flexible queries), gRPC (high performance).

Examples: Stripe API, OpenAI API, Anthropic API, Google Maps API

API Keys & Codes

Tagline: Your identity card for accessing AI services

API keys are unique strings that identify and authenticate you when calling an API. They control access, track usage, and enforce rate limits. **NEVER share your API key publicly.**

Analogy: Like a gym membership card — swipe to get in (authenticate), tracks visits (usage), membership level determines equipment access (permissions).

SDKs & Libraries

Tagline: Pre-built code that makes APIs easy to use

An SDK wraps an API into simple function calls. They handle authentication, retries, error parsing, and response typing automatically.

Examples: openai (Python), @anthropic-ai/sdk, LangChain, Vercel AI SDK

Function / Tool Calling

Tagline: How AI models reach beyond text generation

Function calling is the ability for an LLM to output structured tool calls instead of just text. The model generates a JSON object describing which function to invoke and with what arguments. Your application executes the function and feeds the result back.

Analogy: A manager who writes detailed memos for an assistant to execute. The model is the manager, your code is the assistant.

Connection Map

From	Relationship	To
AI Agents	discover tools via	MCP
AI Agents	call services through	APIs
MCP	wraps and standardizes	APIs
API Keys	authenticate	APIs
SDKs	wraps for easy use	APIs
Function Calling	core capability of	AI Agents
Function Calling	mechanism behind	MCP

7. Code Assistants & AI Editors

10 tools for AI-assisted development — from IDE extensions to CLI agents to cloud IDEs.

GitHub Copilot IDE Extension $10-39/mo

The original AI pair programmer. Inline completions, chat, multi-model support (GPT-5.2, Claude, Gemini). PR summaries and CLI integration.

Platforms: VS Code, JetBrains, Neovim, Visual Studio, Xcode · Learn more

Cursor AI-Native IDE Free / $20/mo

VS Code fork rebuilt for AI. Codebase-wide context with @-mentions, multi-file Composer, agent mode for autonomous execution. All VS Code extensions work.

Platforms: macOS, Windows, Linux · Learn more

Claude Code CLI Agent API usage-based

Agentic CLI tool from Anthropic. Full codebase awareness, multi-file editing, terminal commands, git workflow management, MCP integration, hooks system.

Platforms: Terminal (macOS, Linux, WSL), VS Code, JetBrains · Learn more

Windsurf AI-Native IDE Free / $15/mo

AI IDE (formerly Codeium) with Cascade agent for multi-step task execution, deep codebase indexing, terminal command execution, Supercomplete.

Platforms: macOS, Windows, Linux · Learn more

Sourcegraph Cody IDE Extension Free / $9/mo

AI assistant backed by Sourcegraph's code intelligence. Multi-repo understanding, enterprise-ready with SSO. Custom commands for workflows.

Platforms: VS Code, JetBrains, Neovim, Web · Learn more

Aider CLI Agent Free (open source)

Open-source CLI coding assistant. Works with any LLM, git-aware with auto commits, multi-file editing, voice coding, repo mapping.

Platforms: Terminal (macOS, Linux, Windows) · Learn more

Continue IDE Extension Free (open source)

Open-source autopilot. Connect any LLM (local or API), tab autocomplete, customizable slash commands, fully extensible.

Platforms: VS Code, JetBrains · Learn more

Replit Agent
Cloud IDE + Agent | Free / $25/mo

Browser-based IDE with AI agent. Build full apps from natural language, auto deployment, built-in database and auth. No local setup.

Platforms: Web Browser, Mobile · Learn more

Bolt.new
Browser IDE + Agent | Free / $20/mo

Browser-based AI dev environment by StackBlitz. Full-stack apps via WebContainers, instant previews, one-click Netlify deploy.

Platforms: Web Browser · Learn more

GitHub Copilot Workspace
Task-Oriented Agent | Incl. w/ Copilot

Task-centric environment. Takes a GitHub issue → generates implementation plan → produces multi-file code changes. Iterative refinement.

Platforms: Web (GitHub) · Learn more

8. Resources & Community

30+ curated links for staying current on AI — research, news, forums, newsletters, courses, and video.

Research & Papers

Resource	Description	Link
arXiv	Primary preprint server for AI/ML research. Free access to all papers.	arxiv.org
Papers With Code	Links papers to open-source implementations and benchmarks.	paperswithcode.com
Semantic Scholar	AI-powered academic search engine by Allen Institute.	semanticscholar.org
Google Scholar	Broad academic search. Citation metrics and author profiles.	scholar.google.com

News & Publications

Resource	Description	Link
The Verge — AI	AI coverage for broad audience. Product launches, policy.	theverge.com/ai
TechCrunch — AI	Startup and enterprise AI. Funding, launches.	techcrunch.com/ai
Ars Technica — AI	Technically-informed AI reporting with depth.	arstechnica.com/ai
MIT Technology Review	Premium AI journalism. Research and ethics.	technologyreview.com
AI News	Dedicated AI news outlet. Enterprise focus.	artificialintelligence-news.com

Forums & Communities

Resource	Description	Link
r/MachineLearning	Largest ML research community. Paper discussions, AMAs.	reddit.com/r/ML
r/LocalLLaMA	Running LLMs locally. Quantization, hardware, benchmarks.	reddit.com/r/LocalLLaMA
r/ChatGPT	ChatGPT users. Prompt techniques, use cases.	reddit.com/r/ChatGPT
r/ClaudeAI	Claude community. Tips, API discussions.	reddit.com/r/ClaudeAI
r/Singularity	AGI discussion. Benchmark tracking.	reddit.com/r/singularity
Hacker News	Y Combinator's tech forum. Technical depth.	news.ycombinator.com
Hugging Face	Open-source AI hub. 500K+ models, datasets, spaces.	huggingface.co
Anthropic Discord	Official Anthropic community. Claude API help.	discord.gg/anthropic
OpenAI Forum	Official developer community. API support.	community.openai.com
Latent Space	AI engineering community and podcast.	latent.space

Newsletters

Newsletter	Description	Link
TLDR AI	Daily 5-minute newsletter. Research + industry.	tldr.tech/ai
The Batch — Andrew Ng	Weekly from DeepLearning.AI. Curated picks.	deeplearning.ai/the-batch
Import AI — Jack Clark	Weekly. AI policy, research, capabilities.	importai.substack.com
The Neuron	Daily. Practical focus, tool recommendations.	theneurondaily.com
Ben's Bites	Daily. Product launches, tool discoveries.	bensbites.com

Courses & Learning

Course	Description	Link
fast.ai	Free, world-class deep learning course. Top-down approach.	course.fast.ai
DeepLearning.AI	Andrew Ng's platform. Short courses with AI labs.	deeplearning.ai
Anthropic Docs	Claude API, prompt engineering, Agent SDK, MCP.	docs.anthropic.com
OpenAI Docs	GPT API, function calling, embeddings, fine-tuning.	platform.openai.com/docs
Hugging Face NLP Course	Free NLP with Transformers. Hands-on.	huggingface.co/learn

Podcasts & Video

Channel	Description	Link
Latent Space Podcast	AI engineering. Practitioners from Anthropic, OpenAI.	latent.space/podcast
Lex Fridman Podcast	Long-form interviews with AI leaders.	lexfridman.com
Yannic Kilcher	ML paper explanations and walkthroughs.	YouTube
3Blue1Brown	Visual math + neural network explainers.	YouTube
Two Minute Papers	Short AI research summaries with demos.	YouTube

Staying Current: Recommended Routine

Daily

- Check TLDR AI or The Neuron
- Skim r/LocalLLaMA and r/MachineLearning
- Glance at Hacker News front page

Weekly

- Read The Batch or Import AI
- Listen to one Latent Space episode
- Try one new tool or model

Monthly

- Complete a DeepLearning.AI short course
- Read 1-2 papers via Papers With Code
- Build a small project with a new technique

9. Build Guides

24 hands-on tutorials organized by provider. From beginner chatbots to advanced agents and MCP servers.

Anthropic (Claude) — 6 Guides

#	Guide	Level	Concepts
A1	**Build a Chatbot with Claude**	Beginner	SDK setup, system prompts, conversation memory
A2	**Streaming Responses**	Beginner	Token streaming, event handling, perceived latency
A3	**Tool Use & Function Calling**	Intermediate	Tool definitions, tool use loop, JSON schema
A4	**RAG: Document Q&A**	Intermediate	Document chunking, embeddings, vector stores, ChromaDB
A5	**Building AI Agents**	Advanced	Agent loops, planning, multi-step reasoning, Agent SDK
A6	**Build an MCP Server**	Advanced	MCP protocol, tool registration, resource exposure

OpenAI (GPT) — 6 Guides

#	Guide	Level	Concepts
O1	**Build a Chatbot with GPT**	Beginner	Chat Completions API, message roles, history
O2	**Function Calling with GPT**	Intermediate	Function definitions, tool choice, parallel calls
O3	**RAG with OpenAI Embeddings**	Intermediate	text-embedding-3-small, vector search
O4	**OpenAI Assistants API**	Intermediate	Threads & Runs, file search, code interpreter
O5	**Structured Output**	Intermediate	JSON mode, Pydantic integration, schema enforcement
O6	**Vision & Image Analysis**	Advanced	Vision API, image encoding, multi-image comparison

Google (Gemini) — 6 Guides

#	Guide	Level	Concepts
G1	**Build a Chatbot with Gemini**	Beginner	Google GenAI SDK, chat sessions, safety settings
G2	**Multimodal Input**	Intermediate	Image/PDF/audio analysis, native multimodal
G3	**Image Generation**	Intermediate	Text-to-image, Imagen, style control
G4	**Function Calling**	Intermediate	Function declarations, auto function calling
G5	**Embeddings & Search**	Intermediate	Google embeddings API, cosine similarity
G6	**Grounded Search**	Advanced	Google Search grounding, source citations

Llama (Meta — Run Locally) — 6 Guides

#	Guide	Level	Concepts
L1	**Run Llama Locally with Ollama**	Beginner	Ollama setup, local inference, API compatibility
L2	**Use Llama with OpenAI SDK**	Beginner	Base URL override, provider switching, zero cost
L3	**Local RAG with Llama**	Intermediate	Local embeddings, private RAG, LangChain + Ollama
L4	**Fine-Tune Llama on Your Data**	Advanced	QLoRA fine-tuning, Hugging Face PEFT, model merging
L5	**Local AI Agents**	Advanced	Ollama tool calling, LangChain agents, private automation
L6	**Quantize & Optimize Models**	Advanced	GGUF format, quantization levels, llama.cpp

Quick Start

```
git clone <your-repo-url>
cd mcp_api/guides
pip install -r requirements.txt
cp .env.example .env
# Edit .env and add your API keys:
# ANTHROPIC_API_KEY=sk-ant-...
# OPENAI_API_KEY=sk-...
# GOOGLE_API_KEY=AIza...
python anthropic_01_chatbot.py
```

AI Explorer — Built as an interactive learning experience for understanding AI from the ground up.

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